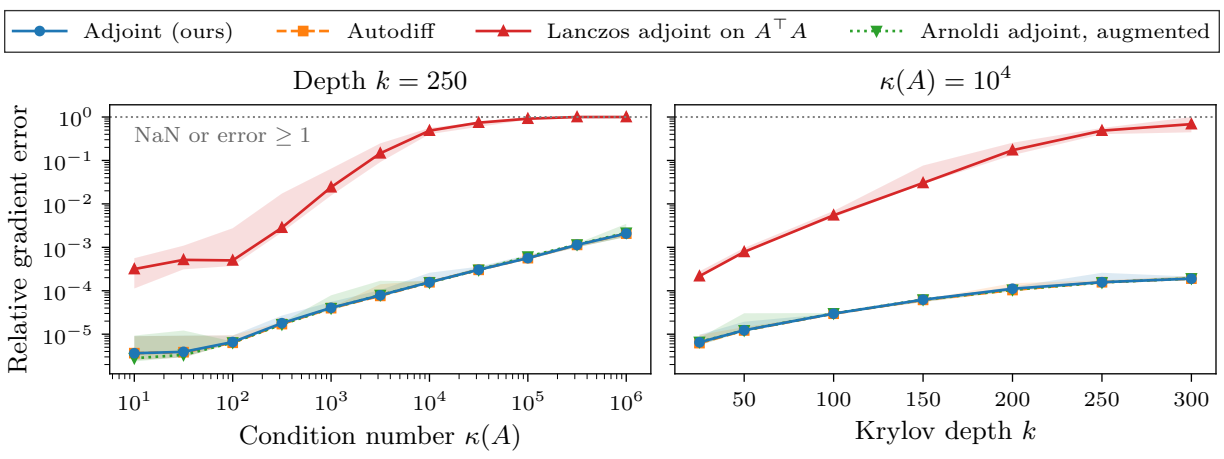

Adjoint (ours)
Autodiff
Lanczos adjoint on $A^\top A$
Arnoldi adjoint, augmented
Depth $k = 250$
$\kappa(A) = 10^4$
Relative gradient error
NaN or error ≥ 1
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^1
10^2
10^3
10^4
10^5
10^6
Condition number $\kappa(A)$
50
100
150
200
250
300
Krylov depth k